

Data Preprocessing Steps and Business Questions with Visualization

Step 1: Data Collection

- ❖ Collected the India's Agricultural Crop Production (1997-2021) dataset.
- ❖ Imported the dataset into Tableau.

Step 2: Data Cleaning

- ❖ Removed duplicate records.
- ❖ Checked and handled missing values.
- ❖ Corrected incorrect data types.
- ❖ Removed unnecessary columns.
- ❖ Standardized state and crop names.

Step 3: Data Transformation

- ❖ Renamed columns for better readability.
- ❖ Converted production, area, and yield columns into numeric format.
- ❖ Formatted the Year field as a date/number.
- ❖ Verified unit values for consistency.

Step 4: Data Validation

- ❖ Checked data accuracy.
- ❖ Verified production, area, and yield values.
- ❖ Ensured there were no inconsistent records.

Step 5: Calculated Fields

- ❖ Created calculated fields such as:
- ❖ Total Production
- ❖ Total Yield
- ❖ Average Yield
- ❖ Total Cultivated Area

- ❖ Production Percentage
- ❖ Rank by Production (optional)

Step 6: Dashboard Preparation Connected the cleaned dataset to Tableau

- ❖ Created worksheets for each visualization.
- ❖ Added filters (Year, State, Crop, Season).
- ❖ Built an interactive dashboard.

Visualization:

Visualization	Chart Used
Total Production & Total Yield	KPI Tiles
Trend of Agricultural Yield (1997-2021)	Line chart
State-wise Agricultural Production Distribution	Filled Map(India Map)
Crop-wise Total Production Analysis	Horizontal Bar Chart
Slate-wise Cultivated Area Comparison	Grouped Bar Chart
Leading Stales by Agricultural Production	Vertical Bar Chart
Year-wise Total Agricultural Production (1997-2021)	Column(Bar) Chart
Production Distribution by Unit Type	Donut Chart
Season-wise Agricultural Production Share	Pie chart

Business Questions

1. What is the total agricultural production between 1997 and 2021?
2. How has agricultural yield changed over the years?

3. Which states contribute the highest agricultural production?
4. Which crops have the highest total production?
5. Which states have the largest cultivated area?
6. How has agricultural production changed year by year?
7. Which unit type contributes the most to total production?
8. What is the production share across different agricultural seasons?
9. Which states consistently perform well in agricultural production?
10. How do cultivated area and production compare across different states?

Expected Insights

1. Identify the leading agricultural producing states.
2. Understand long-term production and yield trends.
3. Compare crop production across different crops.
4. Analyze cultivated area by state.
5. Evaluate seasonal contributions to agricultural production.
6. Identify production patterns from 1997 to 2021.
7. Support data-driven agricultural planning and policy decisions